

Child ASR Adaptation with Adult Retention: An Empirical Study

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Abstract

Automatic Speech Recognition (ASR) systems often underperform for children and non-native speakers, while adapting adult ASR models to child speech can cause adult-speech forgetting. We study *child ASR adaptation with adult retention* across Arabic and English. We compare full fine-tuning, LoRA, and post-hoc weight-space merging across encoder-decoder, encoder-CTC, and AudioLLM-based ASR systems. Experiments use Arabic native and non-native child speech, English MyST child speech, and adult benchmarks from MGB-2 and LibriSpeech test-clean. We evaluate recognition quality with WER and quantify the adaptation-retention trade-off using Retention Index, Child Adaptation Gain, and Adaptation Recovery. Results show that child adaptation is necessary, especially for non-native Arabic and English child speech, but direct adaptation often reduces adult ASR performance. Bilingual adaptation is more stable than language-specific adaptation. Weight-space merging often improves the trade-off, especially for encoder-CTC, Whisper, and AudioLLM-based ASR, with LERP favoring adult retention and TIES recovering stronger child gains. For the encoder-decoder model, direct bilingual fine-tuning remains strongest in raw WER.¹

1 Introduction

Automatic Speech Recognition (ASR) has achieved strong performance on adult speech, but children’s speech remains difficult to recognize. Compared with adults, children often have higher pitch, shorter vocal tracts, greater pronunciation variability, and more disfluencies, all of which can degrade ASR accuracy (Wu et al., 2019; Bhardwaj et al., 2022). The problem is further amplified by the limited availability of labeled child-speech data and by bilingual settings, where models must handle both age-related and language-related shifts.

¹Code, and models are available at <https://github.com/qcri/Child-ASR-Adaptation>.

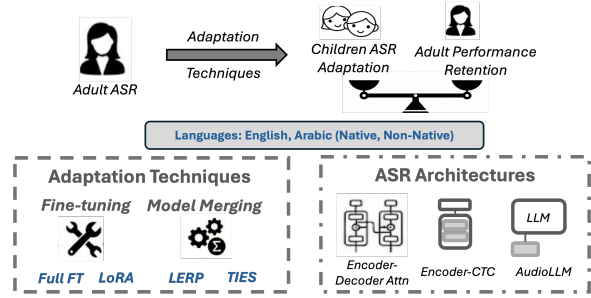


Figure 1: Overview of the child ASR adaptation pipeline across architectures, adaptation methods, and adult-retention evaluation.

A common solution is to adapt adult-trained ASR models using child-speech data. This can improve child recognition on datasets such as MyST (Pradhan et al., 2024) and CMU Kids (Eskenazi et al., 1997), but it can also reduce performance on adult speech due to catastrophic forgetting. This creates a deployment challenge for systems that must serve both adults and children, such as home assistants, classroom tools, and bilingual speech interfaces. We therefore study the *adaptation-retention* trade-off: improving child ASR while preserving adult ASR performance.

This paper presents a retention-aware empirical study of adult-to-child ASR adaptation. We compare three practical adaptation strategies: full fine-tuning, LoRA-based parameter-efficient adaptation (Hu et al., 2021), and post-hoc weight-space merging. For merging, we evaluate Linear Interpolation (LERP) and TIES (Yadav et al., 2023), following the broader motivation of model soups and task arithmetic (Wortsman et al., 2022; Ilharco et al., 2023). Unlike fine-tuning, merging combines adult and child-adapted checkpoints directly in parameter space without additional supervised training, allowing us to test whether child gains can be recovered while reducing adult forgetting. Figure 1 summarizes our experimental pipeline.

We evaluate these strategies across four ASR families: a bilingual encoder–decoder model, a multilingual encoder–decoder model (Whisper Large v3 (Radford et al., 2023)), a bilingual encoder–CTC model, and a multilingual AudioLLM-based ASR system built from Qwen2.5–Omni–7B and adapted for Arabic–English ASR. Our evaluation covers Arabic native and non-native child speech from AraYoungVoices (Chowdhury et al., 2026) (adaptation on ages 7–18; evaluation on its AraKids 7–12 test split), English child speech from MyST (Pradhan et al., 2024), and adult retention benchmarks from LibriSpeech test-clean (Panayotov et al., 2015) and MGB-2 (Ali et al., 2016). In addition to WER, we report Retention Index, Child Adaptation Gain, and Adaptation Recovery to quantify both recognition improvement and retention cost.

Our results reveal several consistent trends. First, child adaptation is necessary across architectures, especially for non-native Arabic children and for English child speech in bilingual settings. Second, language-specific adaptation can cause cross-language degradation, while bilingual adaptation is generally more stable. Third, direct fine-tuning often gives strong child performance gains. However, it can substantially reduce adult ASR performance, particularly on MGB-2. Fourth, weight-space merging often improves the adaptation–retention balance, especially for encoder–CTC, Whisper, and AudioLLM-based ASR systems. LERP tends to preserve adult performance, while TIES more aggressively recovers child-speech gains. For the encoder–decoder model, however, direct bilingual fine-tuning remains strongest in raw WER.

Overall, this study provides practical guidance for building ASR systems that better serve both children and adults. It shows retention-aware evaluation is essential for child ASR adaptation and model merging can be a simple and effective tool for reducing the cost of adult-speech forgetting.

2 Related Work and Positioning

2.1 Child speech recognition and benchmarks

ASR trained predominantly on adult speech degrades on children due to acoustic and linguistic mismatch and limited labeled child data. Recent child-ASR studies have explored diverse modeling approaches, including discrete speech representations, end-to-end neural architectures, and adaptation of large pretrained ASR models to children’s

speech (Dutta et al., 2025; Sukhadia and Chowdhury, 2024; Shivakumar and Narayanan, 2022; Jain et al., 2023). Adult→child adaptation is a pragmatic route to lower child WER, but raises an often under-emphasized, deployment-critical question: how to improve child recognition *without sacrificing adult competence* when one system serves both populations (Bhardwaj et al., 2022; Shivakumar and Georgiou, 2020; Jain et al., 2023; Shankar et al., 2025). Classic corpora (OGI Kids, PF-STAR) and modern benchmarks (CMU Kids read speech, MyST conversational child–agent dialogue) span controlled and conversational settings (Shobaki et al., 2000; Batliner et al., 2005; Eskenazi et al., 1997; Pradhan et al., 2024).

Prior child-ASR research has largely focused on English, with comparatively limited evaluation in other languages. We therefore extend the analysis to Arabic child speech to examine whether adaptation gains and adult-speech retention generalize across language and nativity shifts. Bilingual and L2 settings add further difficulty, as age-related variability interacts with cross-lingual phonology and accent transfer; PF-STAR includes non-native child speech, while TLT-school targets L2 learners in school settings (Batliner et al., 2005; Gretter et al., 2020). For Arabic, child-specific resources remain scarce, and strong pretrained models still show substantial gaps relative to adult speech (Lu et al., 2022; Alkadri et al., 2025). AraYoungVoices provides an Arabic child-adolescent-speech resource covering both native (L1) and non-native (L2) speakers. In this work, we use AraKids (the children subset) as the Arabic data for this task, enabling evaluation under combined age and nativity shift.

2.2 Adult ASR adaptation to Children and the cost downside

Child adaptation mainly targets end-to-end architectures, including encoder–CTC (Graves et al., 2006) and attention-based encoder–decoder models (Chan et al., 2015), often with Conformer-style encoders (Gulati et al., 2020). Large-scale pre-training has produced strong general-purpose backbones (e.g., wav2vec 2.0 and Whisper) adaptable with modest labeled data (Baevski et al., 2020; Radford et al., 2023), and speech-language stacks pairing audio encoders with LLMs have emerged as alternative ASR backbones/decoders (Rubenstein et al., 2023; Tang et al., 2024; Chu et al., 2024). These trends motivate comparing adaptation strate-

gies across *multiple* model families rather than a single architecture. A growing body of work fine-tunes pretrained ASR on child corpora (e.g., MyST, CMU Kids, PF-STAR), including Whisper adaptation studies and variants that control data quality or mix adult/childrenized speech to reduce mismatch (Jain et al., 2023; Attia et al., 2024; Southwell et al., 2024; Graave et al., 2024; Shankar et al., 2025). These studies consistently show large child-domain gains, but evaluation is frequently centered on child-only WER, leaving the *adult regression* induced by child adaptation less systematically quantified. When one system must serve both adults and children, child-only WER can look like progress while masking a real adult regression that the same system still has to absorb. This motivates retention-aware evaluation where adult ASR performance is explicitly anchored as a first-class constraint. Sequential adaptation induces catastrophic forgetting, where optimizing for a new domain degrades previously learned ones. Continual learning mitigates this via regularization (e.g., EWC), distillation-style constraints (e.g., Learning without Forgetting), and evaluation constructs such as forgetting/backward transfer computed from task-wise performance matrices (Kirkpatrick et al., 2017; Li and Hoiem, 2017; Lopez-Paz and Ranzato, 2017; Chaudhry et al., 2018). In speech recognition, continual learning and domain expansion studies similarly report old-domain degradation and explore constraints for speech models (Hadian and Gorin, 2021). Our study operationalizes this perspective with adult anchors and retention-focused metrics (ratio-based retention/recovery scores), enabling direct comparison of adaptation strategies under an explicit adult-performance requirement.

2.3 Parameter-efficient adaptation and weight-space merging

Parameter-efficient fine-tuning (PEFT) adapts large pretrained backbones while updating only a small number of parameters, reducing training and storage cost. Adapters and LoRA are widely used in NLP (Houlsby et al., 2019; Hu et al., 2021) and have recently been explored for adapting speech foundation models (e.g., Whisper) in multilingual settings (Song et al., 2024). Related child-ASR work also studies lightweight modules and constrained updates to mitigate mismatch in self-supervised representations (Fan and Alwan, 2022). We include LoRA as a *practitioner-standard* baseline and compare its child gains and adult regres-

sion against full fine-tuning and post-hoc weight-space merging under the same adult anchors and retention metrics across ASR architectures, framing the study as a controlled empirical comparison rather than a new adaptation method. Weight-space model merging offers a complementary, post-hoc alternative: combining an adult anchor and a child specialist without additional gradient-based training. Model soups average across multiple fine-tuned checkpoints, while task arithmetic interprets fine-tuning as task vectors composed by addition/subtraction (Wortsman et al., 2022; Ilharco et al., 2023). TIES-Merging further addresses destructive interference by pruning small-magnitude updates, resolving sign conflicts, and selectively merging aligned parameters (Yadav et al., 2023). These methods are attractive for *adult-retentive* child adaptation because they can recover child gains while preserving adult performance relative to naive child-only fine-tuning.

2.4 Positioning this work

Building on prior child ASR adaptation studies and continual-learning insights on forgetting, we frame *child ASR adaptation with adult retention* as a retention-aware empirical comparison rather than a new algorithmic proposal. We systematically evaluate (i) full fine-tuning, (ii) LoRA-based PEFT, and (iii) post-hoc weight-space merging (LERP and TIES) across encoder-decoder attention models, encoder-CTC models, Whisper, and an AudioLLM-based ASR. We anchor adult performance on LibriSpeech test-clean and MGB-2 and extend child evaluation beyond English with AraYoungVoices (the test set is Arabic L1/L2, 7–12 years-old age group, referenced to as AraKids) and MyST, enabling a direct characterization of adaptation-retention trade-offs across architectures and languages (Jain et al., 2023; Shankar et al., 2025; Fan and Alwan, 2022; Yadav et al., 2023; Pradhan et al., 2024; Radford et al., 2023). The contribution is therefore empirical rather than algorithmic. We study adult-to-child adaptation under combined age, language, and nativity shifts, and we show that the trade-off between child-speech improvement and adult-speech retention depends on the architecture.

3 Datasets and Benchmarking

We adapt models using child-speech data and evaluate child WER on held-out child test sets. Adult-

Domain	Dataset	Trn	Dev	Tst	Role
Child	AraYoungVoices (Ar)	✓	✓	✓	Adapt + child eval
Child	MyST (En)	✓	✓	✓	Adapt + child eval
Adult	MGB-2 (Ar)	–	✓	✓	Adult retention
Adult	LibriSpeech (En)	–	✓	✓	Adult retention

Table 1: Datasets and split usage. Child data is used for adaptation and child evaluation, while adult data is used only for retention evaluation. For Arabic, training and development data come from the full AraYoungVoices corpus (ages 7–18), while child evaluation uses the AraKids USUP test set (ages 7–12).

Dataset	#Spk	#Utt	Hours
<i>Training data</i>			
AraYoungVoices total	199	≈62K	102.94
AraYoungVoices L1	107	≈32K	45.40
AraYoungVoices L2	92	≈329K	57.54
MyST	–	≈77K	171.20
Total	–	≈141K	280.38
<i>Test data</i>			
AraKids total	66	2K	4.00
AraKids L1	26	1K	1.57
AraKids L2	40	1K	2.43
MyST	–	13,180	28.32
Total	–	15,180	32.32

Table 2: Child adaptation data overview. Training rows report the AraYoungVoices corpus (ages 7–18), split into native (L1) and non-native (L2) speakers; test rows report the AraKids USUP test set (ages 7–12), which is the Arabic child evaluation set used throughout the paper. For MyST, we use the official split.

Group	Train			Test	
	#Spk	#Utt	Hours	#Spk	#Utt
<i>L1 (native)</i>					
Egyptian	33	9,906	14.45	5	252
Gulf	25	7,429	10.95	7	249
Levantine	25	7,089	8.51	7	250
North African	24	7,853	11.50	7	249
<i>L2 (non-native)</i>					
Africa	13	4,242	9.23	7	250
Americas	14	4,582	10.01	5	250
Asia	30	9,449	18.26	15	250
Europe	35	11,024	20.04	13	250
Total	199	61,574	102.94	66	2,000

Table 3: AraYoungVoices distribution by group. L1 groups are Arabic macro-regions; L2 groups are the speakers’ regions of origin. Train and test speakers are disjoint. Train columns report the AraYoungVoices training set (ages 7–18); test columns report the AraKids USUP test set (ages 7–12).

speech datasets only serve as retention anchors to measure how much adult ASR performance is preserved after child adaptation (Tables 1, 2).

We use AraYoungVoices (Chowdhury et al., 2026) and MyST (Pradhan et al., 2024) as our primary datasets for adapting, and testing. Hyperparameters are selected on the development sets, and

child ASR performance is reported on held-out test sets.

AraYoungVoices AraYoungVoices is an Arabic children and adolescent read-speech corpus with 61,574 training utterances from 199 speakers, totaling 102,94 hours. It contains two subsets: *L1*, covering native Arabic speakers from Egyptian, Gulf, Levantine, and North African regions, and *L2*, covering non-native Arabic speakers from Africa, Asia, the Americas, and Europe with at least 13 unique speakers from each reported group. Prompts are educator-designed and fully diacritized. L2 prompts are adapted for different proficiency levels from government-certified books, while controlling lexical and syntactic complexity. The AraYoungVoices training set includes speakers aged 7–18 years, while the test set used (AraKids) includes children aged 7–12 years. AraKids is released with two reference types: a *prompt transcription* (the written sentence the child was asked to read), and a *verbatim transcription* produced by human annotators, which records hesitations, repetitions, fillers, and dialectal spellings in place of Modern Standard Arabic forms. We score against the prompt transcription throughout, so the reported Arabic child WER measures agreement with the prompt transcription rather than with a verbatim transcription. Table 3 reports the number of speakers and utterances in each of the eight L1/L2 groups, for both the training and the test split. The AraKids test splits are speaker- and prompt-disjoint (USUP: Unseen Speaker Unseen Prompt), which means no prompt/speaker overlap between train and test. The test set is group-balanced, with approximately 250 utterances sampled from each of the 8 groups: 1,000 utterances for AraKidsL1 and 1,000 for AraKidsL2 yielding a total of 2000 test utterances from 66 speakers of 3.99 hours². The test set is balanced by utterance count across the eight reported L1/L2 groups; speaker counts are not necessarily equal across groups.

MyST We use MyST (Pradhan et al., 2024), an English child-speech corpus of conversational tutoring dialogues, following its official train, development, and test splits. MyST includes students from Grades 3, 4, and 5; the source paper reports grade levels rather than an exact age range.

²The average utterance duration is 5.64s for AraKidsL1 and 8.74s for AraKidsL2.

Adult Test sets Adult retention is evaluated using Arabic MGB-2 (Ali et al., 2016) and English LibriSpeech test-clean (Libri-C) (Panayotov et al., 2015). These adult datasets are excluded from child adaptation and are used only to measure whether adapted models preserve performance on adult speech.

4 Methodology

We study child ASR adaptation under an adult-retention constraint. Given an adult-pretrained ASR model, the goal is to improve recognition on child speech while preserving the model’s original performance on adult speech. We compare three adaptation paradigms: full fine-tuning, parameter-efficient fine-tuning with LoRA, and post-hoc weight-space merging.

Full fine-tuning and LoRA directly adapt the base model using child-speech data. In contrast, weight-space merging is applied after the adaptation where it combines child-adapted checkpoints with the original model to obtain a single model that balances child-speech improvement and adult-speech retention. Since full fine-tuning and LoRA follow standard training procedures, their implementation details are given in Section 5.2. In this section, we focus on the merging methods and the evaluation metrics used to quantify adaptation, retention, and recovery.

Post-hoc merging Let θ_{base} denote the shared base model and let $\theta^1, \dots, \theta^N$ denote N child-specialized checkpoints that are parameter-compatible with θ_{base} . Each θ^i is a child-specialized checkpoint produced by adapting the same adult-pretrained base model. Depending on the experiment, the base model is adapted using Arabic child speech, English child speech, or the combined Arabic–English child data through full fine-tuning or LoRA. All merging operations are performed only within a single architecture family (encoder–decoder or encoder–CTC among others) and between checkpoints sharing the same initialization.

Linear interpolation (LERP) We apply linear interpolation in weight space, which takes a weighted average of model parameters. Given the merge set $\{\theta^1, \dots, \theta^N\}$ (which includes the adult base θ_{base}), the merged model is: $\theta_m = \sum_{i=1}^N \alpha_i \theta^i$, $\alpha_i \in [0, 1]$ $\sum_{i=1}^N \alpha_i = 1$. Note, that unlike task-arithmetic methods, LERP operates di-

rectly on the raw weights and does not construct task vectors relative to θ_{base} .

TIES For each child-adapted checkpoint θ^i , we define its *task vector* as $\tau^i = \theta^i - \theta_{base}$, that is, the parameter update introduced by child adaptation relative to the shared adult base model. TIES (Yadav et al., 2023) first measures how each child-adapted model differs from the shared adult base model. It removes very small parameter changes, which are assumed to have limited importance. When different adapted models change the same parameter in conflicting directions, TIES selects the direction supported by the majority of models. It then averages only the updates that agree with this direction and adds the resulting update to the base model. In contrast to LERP, which averages raw weights, TIES operates on task vectors and resolves sign conflicts before averaging.

Selective merging for encoder–decoder models

We also explore selective merging for encoder–decoder ASR models. The goal is to control which components are merged in order to improve child adaptation and adult retention. In addition to full-model merging, we evaluate an encoder-only variant. We merge only the encoder parameters and keep the decoder fixed to the base (adult) checkpoint. This choice is motivated to analyze the contribution of the encoder and decoder components in model merging. In encoder–decoder ASR architectures, the encoder primarily learns acoustic representations, while the decoder models output token dependencies conditioned on these representations. By keeping the decoder fixed to the adult checkpoint, we isolate the effect of merging updates in the encoder and evaluate whether most of the beneficial adaptation is concentrated in the acoustic representation space. This setup also allows us to assess the impact of restricting merging to a subset of parameters, which may reduce interference between task-specific updates.

CTC head anchoring for encoder–CTC models

For encoder–CTC models, we merge encoder parameters but keep the adult CTC projection head fixed. We also tested swapping child-specialized CTC heads trained on individual AraYoungVoices, MyST, and joint AraYoungVoices+MyST.

5 Experimental Setup

5.1 ASR Model Architectures

We evaluate child ASR adaptation across four ASR families: bilingual encoder–decoder, multilingual Whisper, bilingual encoder–CTC, and AudioLLM-based ASR.

Bilingual Encoder–Decoder Model (ED-Bi)

ED-Bi is an adult-pretrained Arabic–English Conformer–Transformer ASR model (Chowdhury et al., 2021). It uses a self-supervised speech front-end, a 12-block Conformer encoder, and a 6-block Transformer decoder. The model follows a hybrid CTC–attention objective with $\lambda_{\text{CTC}} = 0.3$.

Whisper Model (ED-MI) ED-MI is Whisper Large v3, a multilingual encoder–decoder ASR model pretrained on large-scale multilingual and multitask speech data (Radford et al., 2023). We use it as a high-capacity reference model for adaptation and retention analysis.

Bilingual Encoder–CTC Model (E-CTC-Bi) E-CTC-Bi is an encoder-only Arabic–English ASR model (FANAR Team et al., 2026) based on OWSM-CTC (Peng et al., 2024). It uses a frozen Arabic-centric self-supervised front-end (Sukhadia and Chowdhury, 2026), Transformer encoder layers, and self-conditioned CTC with intermediate and final-layer CTC losses.

AudioLLM-based ASR Model (ALLM-ASR-Bi) ALLM-ASR-Bi uses *Qwen2.5-Omni-Ar-En-7B* (Bhatti et al., 2026)³, an Arabic–English ASR-adapted variant of Qwen2.5-Omni-7B (Qwen Team, 2025). It combines audio understanding with instruction-style decoding. All child-adaptation experiments start from this bilingual ASR-oriented checkpoint.

5.2 Child Adaptation Configuration

We compare full fine-tuning, LoRA adaptation, and post-hoc model merging. All checkpoints are selected using development-set WER on AraYOUNGVoices and MyST.

Full fine-tuning. ED-Bi and E-CTC-Bi are adapted for 5 epochs, while ED-MI is adapted for 1 epoch to reduce overfitting. For E-CTC-Bi, we use AdamW with peak learning rate 2.5×10^{-4} , piecewise linear warm-up, and intermediate CTC losses at layers $\{6, 9, 11\}$; the SSL front-end is frozen.

For ED-Bi, we use the hybrid CTC–attention objective, Adam with learning rate 2×10^{-4} , and 15k warm-up steps; the XLSR-53 front-end is frozen. For ED-MI, we use learning rate 10^{-5} , and 500 warm-up steps.

LoRA adaptation. LoRA keeps the base model frozen and updates only low-rank adapters. For ED-Bi and E-CTC-Bi, adapters are inserted into attention projection layers with rank $r = 16$, $\alpha = 16$, and dropout 0.05, and optimized for 5 epochs with AdamW at learning rate 10^{-3} . For ALLM-ASR-Bi, LoRA starts from the bilingual ASR checkpoint, uses rank $r = 16$, $\alpha = 32$, dropout 0.05, and targets all-linear modules with bf16 training, cosine scheduling, learning rate 10^{-4} , and warm-up ratio 0.05.

Model merging. Model merging combines the adult base model θ_{base} with child-adapted checkpoints obtained through full fine-tuning $\theta_{\text{child}}^{\text{ft}}$ or LoRA adaptation $\theta_{\text{child}}^{\text{loRa}}$. We consider both Arabic-only and Arabic–English child-adapted checkpoints. For ALLM-ASR-Bi, the merge pool additionally includes the original Qwen2.5-Omni-7B checkpoint as a multilingual prior. For E-CTC-Bi, we retain the adult CTC output head after merging, since it gives lower validation WER than the child-adapted CTC heads from either the Arabic-only or Arabic–English checkpoints (Figure A.2). For LERP, we sweep $\alpha \in [0.2, 0.8]$ and select $\alpha_1 = 0.2$, $\alpha_2 = \alpha_3 = 0.4$ based on validation-set WER. For TIES, we similarly tune the merge weight w and density d on the validation set. The selected merged models are then evaluated on both child and adult benchmarks to quantify the adaptation–retention trade-off. Figures A.1 and A.2 in the Appendix show the merge-parameter sweeps.

5.3 Evaluation Measures

We use word error rate, $\text{WER}(\theta, D_*)$, as the primary ASR metric. Before scoring, we apply lightweight normalization to both references and hypotheses: Arabic diacritics are removed, common Arabic letter variants are normalized, including *alef*, *hamza*, *alef maksura* $\rightarrow$ *ya*, and *teh marbuta*, and Latin text is lowercased. For MyST, we exclude segments with empty references, non-speech-only markers such as $\langle \text{silence} \rangle$ and $\langle \text{noise} \rangle$, or unintelligible placeholders such as $\langle () \rangle$. Empty hypotheses are retained when the reference contains speech, since they reflect recog-

³Model obtained upon request from the authors.

dition errors. This reduces scoring artifacts from orthographic variation and non-speech regions.

WER remains the primary evaluation metric and is reported for all child- and adult-speech benchmarks. The measures below do not replace WER; they summarize different aspects of the adaptation–retention trade-off and are computed directly from the reported WER values. Following this, we quantify the trade-offs inherent in child ASR adaptation using three complementary metrics. These metrics separate adult-speech retention, child-speech improvement, and the extent to which merged models recover the gains of child-specific fine-tuning:

- *Retention Index* ($\mathcal{RI}$), which measures how well an adapted or merged model preserves adult ASR performance relative to the base model.
- *Child Adaptation Gain* ($\mathcal{AG}$), which measures how much an adapted model improves over the base model on child speech.
- *Adaptation Recovery* ($\mathcal{AR}$), which measures how close a merged model is to the child-specific fine-tuned model.

For each metric, we also report the *signed WER-point change*, where positive values indicate improvement, while negative values indicate degradation relative to the reference model used by that metric.

Retention Index The Retention Index measures how much adult-speech performance is preserved after adaptation:

$$\mathcal{RI}(\theta) = \frac{\text{WER}(\theta_{base}, D_{adult}^L)}{\text{WER}(\theta, D_{adult}^L)}. \quad (1)$$

Here, $L \in \{\text{Ar}, \text{En}\}$, θ_{base} is the original adult ASR model, and θ is the adapted or merged model.

A value of $\mathcal{RI} = 1.0$ indicates that adult performance is preserved. Values < 1.0 indicate forgetting, while values > 1.0 indicate improved adult performance. To make the magnitude of retention gain or loss explicit, we also report the signed WER-point:

$$\Delta_{adult}(\theta) = \text{WER}(\theta_{base}, D_{adult}^L) - \text{WER}(\theta, D_{adult}^L). \quad (2)$$

Positive values indicate adult WER improvement, while negative values indicate adult WER degradation.

Child Adaptation Gain Child Adaptation Gain measures the relative WER reduction of an adapted model over the base model on child speech:

$$\mathcal{AG}(\theta) = \frac{\text{WER}(\theta_{base}, D_{child}^L) - \text{WER}(\theta, D_{child}^L)}{\text{WER}(\theta_{base}, D_{child}^L)}. \quad (3)$$

A value of $\mathcal{AG} = 0$ means that the adapted model performs the same as the base model on child speech. Positive values indicate improvement over the base model, while negative values indicate degradation.

The absolute magnitude of the gain or loss explicit are reported using:

$$\Delta_{child}(\theta) = \text{WER}(\theta_{base}, D_{child}^L) - \text{WER}(\theta, D_{child}^L). \quad (4)$$

Positive values indicate improvement, while negative values indicate that adaptation hurts child ASR.

Adaptation Recovery For merged models, Adaptation Recovery measures how much of the child-specific adaptation gain is recovered by the merged model:

$$\mathcal{AR}(\theta_m) = \frac{\text{WER}(\theta_{base}, D_{child}^L) - \text{WER}(\theta_m, D_{child}^L)}{\text{WER}(\theta_{base}, D_{child}^L) - \text{WER}(\theta_{ft}, D_{child}^L)}. \quad (5)$$

Here, θ_{ft} is the best child-specific adapted model, obtained either through full fine-tuning or LoRA adaptation, and θ_m is the merged model. This metric is computed when θ_{ft} improves over θ_{base} on child speech.

A value of $\mathcal{AR} = 1$ means that the merged model matches the child-specific adapted model. Values within $[0, 1)$ indicate partial recovery of the child-specific adaptation gain, and above 1 indicate that the merged model surpasses the child-specific adapted model, and values < 0 indicate that the merged model performs worse than the base model on child speech. We also report the remaining WER gap to the child-specific adapted model:

$$\Delta_{rec}(\theta_m) = \text{WER}(\theta_{ft}, D_{child}^L) - \text{WER}(\theta_m, D_{child}^L). \quad (6)$$

Positive values indicate that the merged model outperforms the child-specific adapted model, while negative values indicate that it remains worse.

6 Results

Child Adaptation and Adult Retention with Fine-tuning Table 4 reports WER for the adult base models and child-adapted checkpoints.

Encoder-Decoder (ED-Bi)							Encoder-CTC (E-CTC-Bi)						
Model	ArL1	ArL2	MyST	Libri-C	MGB2	Avg.	Model	ArL1	ArL2	MyST	Libri-C	MGB2	Avg.
θ_{base}	9.14	44.11	36.68	9.86	10.92	22.142	θ_{base}	14.72	63.37	26.63	3.83	13.18	24.346
$\theta_{child}^{Ar}(ft)$	6.41	26.76	44.23	12.21	10.70	20.062	$\theta_{child}^{Ar}(ft)$	13.11	16.32	77.32	18.64	23.63	29.804
$\theta_{child}^{En}(ft)$	25.75	106.98	22.68	6.66	14.16	35.246	$\theta_{child}^{En}(ft)$	51.44	92.73	15.11	6.80	26.26	38.468
$\theta_{child}^{Ar+En}(ft)$	6.37	22.71	22.52	6.66	12.97	14.246	$\theta_{child}^{Ar+En}(ft)$	12.45	24.24	15.90	6.66	21.20	16.090
$\theta_{child}^{Ar+En}(l)$	7.16	26.49	23.49	7.12	11.99	15.250	$\theta_{child}^{Ar+En}(l)$	15.16	37.65	18.57	5.91	16.77	18.810
Whisper Large v3 (ED-MI)							Qwen2.5-Omni-7B-Ar-En (ALLM-ASR-Bi)						
Model	ArL1	ArL2	MyST	Libri-C	MGB2	Avg.	Model	ArL1	ArL2	MyST	Libri-C	MGB2	Avg.
θ_{base}	9.68	40.68	21.96	2.47	15.15	17.988	Qwen2.5O	24.24	73.53	15.64	4.07	34.24	30.344
$\theta_{child}^{Ar+En}(ft)$	8.70	11.84	14.47	3.11	23.52	12.328	θ_{base}	6.96	34.58	35.87	1.48	13.27	18.432
							$\theta_{child}^{Ar+En}(l)$	6.50	26.72	15.55	1.56	16.83	13.432

Table 4: Reported WER (%) for the adult θ_{base} model and adapted ASR models across four architectures. ArL1: AraKidsL1, ArL2: AraKidsL2, ft: full fine-tuning, l: LoRA and Qwen2.5: Base weights for the Qwen Omni model (Qwen Team, 2025). Blue marks the best score within each architecture block. Lower WER is better.

MOp	ArL1	ArL2	MyST	Libri-C	MGB2	Avg. WER
Encoder-Decoder (ED-Bi)						
LERP	7.12	27.01	30.43	09.06	11.00	16.924
LERP (Enc)	7.22	27.38	31.26	8.65	11.00	17.102
TIES	7.14	26.76	28.67	9.23	11.26	16.612
TIES (Enc)	7.35	27.01	28.25	8.16	10.86	16.326
Encoder-CTC (E-CTC-Bi)						
LERP	11.20	23.08	20.25	4.37	14.63	14.706
TIES	11.07	18.94	18.68	5.66	18.71	14.612
Whisper-Large v3 (ED-MI)						
LERP	6.55	20.01	17.78	2.43	14.49	12.252
LERP (Enc)	7.64	26.38	20.45	2.41	15.00	14.376
TIES	8.48	11.37	14.36	2.91	22.68	11.96
TIES (Enc)	7.16	19.83	17.55	2.61	17.01	12.832
Qwen2.5-Omni-7B (ALLM-ASR-Bi)						
LERP	6.47	27.01	12.79	1.59	14.33	12.438
TIES	6.44	27.15	12.78	1.60	14.19	12.432

Table 5: Reported WER (%) for different merge operations. Best scores within each architecture are highlighted in blue. MOp: merge operation.

Across architectures, child adaptation improves recognition on child speech, with the largest gains seen on ArL2 and MyST. For ED-Bi, bilingual child adaptation reduces WER by 49% and 38.6% on ArL2 and MyST respectively, giving the best average WER within the ED block. For ED-MI, bilingual adaptation gives even larger gains on ArL2. For ALLM-ASR-Bi, child LoRA improves over all child benchmarks relative to the bilingual adult base, most notably the LoRA is able to recover MyST, where WER drops from 35.87 to 15.55.

The results also show that language-specific adaptation can be unstable. In E-CTC-Bi, Arabic-only adaptation gives the best ArL2 WER, but severely degrades MyST. English-only adaptation gives the best MyST WER, but substantially hurts

Arabic child performance. Bilingual adaptation is therefore more balanced, giving the best average WER for both ED-Bi and E-CTC-Bi.

However, child adaptation is not retention-neutral. Adult performance often degrades after child adaptation, especially on MGB-2. For example, MGB-2 WER increases for all architectures under bilingual child adaptation. This confirms the need to evaluate child ASR adaptation together with adult-speech retention.

Merging and Adaptation-Retention Trade-offs

Tables 5 and 6 (along with Table 7 in Appendix) show that weight-space merging often improves the adaptation-retention balance, but its effect depends on the architecture and merge operation. Direct child adaptation usually gives strong child-speech gains, but it can also reduce adult ASR performance, especially on Arabic. Merging helps to recover child gain from fine-tuning while reducing the adult-retention cost.

For ED-Bi, merging gives modest benefits. The best merged variant is encoder-level TIES, with the lowest average WER for the ED-Bi. However, merged models do not fully match the child WER of bilingual fine-tuning. Their main advantage is retention. LERP keeps MGB-2 close to the base model with $\mathcal{RI} = 0.99$ and only -0.08 WER loss, while TIES gives $\mathcal{RI} = 0.97$. Thus, for ED-Bi, merging is useful for retention, but direct bilingual fine-tuning remains stronger for child WER.

For E-CTC-Bi, merging provides a clearer trade-off improvement. Full fine-tuning gives strong child gains but causes substantial adult forgetting, with $\mathcal{RI} = 0.58$ on Libri-C and $\mathcal{RI} = 0.62$ on

MGB-2. LERP improves retention to $\mathcal{RI} = 0.88$ and 0.90, respectively, while still improving child WER. TIES gives the strongest child recovery, exceeding the child-specific reference on ArL1 and ArL2, but with weaker adult retention than LERP. This shows that LERP is more retention-preserving, whereas TIES is more child-adaptation-oriented.

For *ED-MI*, the same pattern holds. LERP is the safest merge operation for adult retention, reaching $\mathcal{RI} > 1.0$ on both Libri-C and MGB-2, while still improving child WER over the base model. TIES gives stronger child recovery, matching or slightly surpassing the child-specific reference on ArL2 and MyST, but it incurs a larger MGB-2 retention cost.

For *ALLM-ASR-Bi*, merging gives the strongest overall benefit. Both LERP and TIES improve over the child LoRA checkpoint, and TIES gives the best average WER. It improves MyST WER from 15.55 to 12.78 and reduces MGB-2 WER from 16.83 to 14.19. In the trade-off metrics, TIES also gives the best MyST adaptation gain, $\mathcal{AG} = 0.644$ (a 64.4% relative WER reduction), and improves MGB-2 retention to $\mathcal{RI} = 0.94$.

Trends Overall, the results show three trends. First, child ASR adaptation is necessary, especially for non-native Arabic children (ArL2) and English (MyST) data in bilingual modeling setup. Second, direct child adaptation can substantially reduce adult ASR performance, making retention-aware evaluation essential. Third, weight-space merging often provides a better compromise, especially for E-CTC-Bi, ED-MI, and ALLM-ASR-Bi. LERP generally favors adult retention, while TIES more aggressively recovers child-speech gains and can, in some settings, improve both child and adult ASR, as reported above for ALLM-ASR-Bi. This recurring pattern across multiple ASR architectures is one of the central findings of this study.

7 Conclusion

We presented a retention-aware study of adult-to-child ASR adaptation across encoder-decoder, encoder-CTC, Whisper, and AudioLLM-based ASR systems. Using Arabic native and non-native child speech, English MyST, and adult benchmarks from MGB-2 and LibriSpeech test-clean, we evaluated full fine-tuning, LoRA, and post-hoc weight-space merging with WER, Retention Index, and Adaptation Recovery. The results show that child adaptation is necessary, but direct adaptation can reduce adult ASR performance. Bilingual adapta-

Model	$\mathcal{RI} (\Delta_{adult})$		$\mathcal{AR} (\Delta_{rec})$		
	Libri-C	MGB2	ArL1	ArL2	MyST
<i>Encoder-Decoder (ED-Bi)</i>					
FT	1.48 (+3.20)	0.84 (-2.05)	—	—	—
LoRA	1.38 (+2.74)	0.91 (-1.07)	0.715 (-0.79)	0.823 (-3.78)	0.931 (-0.97)
LERP	1.09 (+0.80)	0.99 (-0.08)	0.729 (-0.75)	0.799 (-4.30)	0.441 (-7.91)
TIES	1.07 (+0.63)	0.97 (-0.34)	0.722 (-0.77)	0.811 (-4.05)	0.566 (-6.15)
<i>Encoder-CTC (E-CTC-Bi)</i>					
FT	0.58 (-2.83)	0.62 (-8.02)	—	—	—
LoRA	0.65 (-2.08)	0.79 (-3.59)	-0.194 (-2.71)	0.657 (-13.41)	0.751 (-2.67)
LERP	0.88 (-0.54)	0.90 (-1.45)	1.551 (+1.25)	1.030 (+1.16)	0.595 (-4.35)
TIES	0.68 (-1.83)	0.70 (-5.53)	1.608 (+1.38)	1.135 (+5.30)	0.741 (-2.78)
<i>Whisper Large v3 (ED-MI)</i>					
FT	0.79 (-0.64)	0.64 (-8.37)	—	—	—
LERP	1.02 (+0.04)	1.05 (+0.66)	3.194 (+2.15)	0.717 (-8.17)	0.558 (-3.31)
TIES	0.85 (-0.44)	0.67 (-7.53)	1.224 (+0.22)	1.016 (+0.47)	1.015 (+0.11)
<i>Qwen2.5-Omni-7B (ALLM-ASR-Bi)</i>					
LoRA	0.95 (-0.08)	0.79 (-3.56)	—	—	—
LERP	0.93 (-0.11)	0.93 (-1.06)	1.065 (+0.03)	0.963 (-0.29)	1.136 (+2.76)
TIES	0.92 (-0.12)	0.94 (-0.92)	1.130 (+0.06)	0.945 (-0.43)	1.136 (+2.77)

Table 6: Retention and adaptation recovery trade-offs per benchmark. $\mathcal{RI}$ (adult sets) and $\mathcal{AR}$ (child sets) are reported with their signed WER point change (Δ_{adult} , Δ_{rec}); higher values are better. Reference models are on a gray background, and the best Δ is in blue.

tion is generally more stable than language-specific adaptation. Weight-space merging often improves the adaptation-retention balance, particularly for encoder-CTC, Whisper, and AudioLLM-based ASR models, while direct bilingual fine-tuning remains strongest for the Ar-En encoder-decoder model in raw WER. Overall, our findings highlight model merging as a simple and effective strategy for building ASR systems that better serve both child and adult speakers.

8 Limitations

This study focuses on a retention-aware comparison of widely used adaptation strategies rather than proposing a new ASR architecture or merging algorithm. This choice allows us to isolate how full fine-tuning, LoRA, and post-hoc merging behave across different ASR families, but it also leaves room for more specialized adaptation methods, such as continual-learning objectives, rehearsal-based training, or retention-aware regularization. Our evaluation covers Arabic and English child speech, with Arabic further divided into native and non-native speakers. While this provides a useful bilingual and cross-nativity setting, the results may not fully generalize to other languages, spontaneous child speech, or highly noisy real-world environments. Future work can extend the analysis to broader ranges, more dialects, ad-

ditional speaking styles, and deployment-oriented conditions such as classrooms, homes, and mobile recordings. Broad regional categories may also mask country-specific and dialect-specific variation, so these results should not be interpreted as country-level performance estimates. Speaker-age classification and model routing are valid alternatives, but they address a different system design. Routing requires an additional age classifier and multiple models, which can increase latency, deployment cost, and error propagation, especially when adults and children speak in the same session. A direct comparison between routing and unified adaptation is left for future work.

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A Appendix A.

A.1 Parameter sweep ablation

We ablate the TIES merging configuration along two axes: the per-expert weights/densities (Fig. A.1) and the choice of CTC head (Fig. A.2).

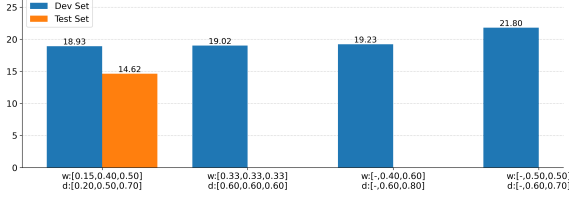


Figure A.1: TIES merging weight/density sweep (average WER, %). Each bar group merges the base model M_0 with expert checkpoints A, B, C using per-expert weights w and densities d , reported in fixed order $[A, B, C]$, where $A = \theta_{child}^{En}$, $B = \theta_{child}^{Ar}$, and $C = \theta_{child}^{Ar+En}$; all variants retain the adult CTC head. Dev (blue) and Test (orange) WER are shown. The selected configuration $w=[0.15, 0.40, 0.50], d=[0.20, 0.50, 0.70]$ attains the lowest WER.

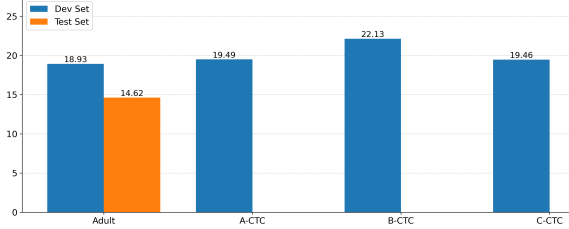


Figure A.2: Effect of CTC head swapping on average WER (%), with the TIES merge fixed to the selected configuration ($w=[0.15, 0.40, 0.50], d=[0.20, 0.50, 0.70]$). *Adult* uses the base adult CTC head; *A-CTC/B-CTC/C-CTC* substitute the CTC head from expert $A/B/C$ while leaving the merged encoder unchanged. The adult head remains best on Dev.

A.2 Adaptation Gain interpretation

Table 7 reports Child Adaptation Gain ($\mathcal{AG}$) per benchmark, complementing the retention and recovery view in Table 6. The relative ordering of methods is architecture-dependent. For ED, full fine-tuning yields the largest child gains on all three child sets ($\mathcal{AG} = 0.303, 0.485, 0.386$ on AraKidsL1, AraKidsL2, and MyST); LoRA and both merge operations give smaller gains, with the gap widest on MyST ($\mathcal{AG}$ dropping to 0.170–0.218 for merging).

Model	$\mathcal{AG} (\Delta_{child})$		
	ArL1	ArL2	MyST
<i>Encoder-Decoder (ED-Bi)</i>			
<i>FT (ref)</i>	0.303 (+2.77)	0.485 (+21.40)	0.386 (+14.16)
LoRA	0.217 (+1.98)	0.399 (+17.62)	0.360 (+13.19)
LERP	0.221 (+2.02)	0.388 (+17.10)	0.170 (+6.25)
TIES	0.219 (+2.00)	0.393 (+17.35)	0.218 (+8.01)
<i>Encoder-CTC (E-CTC-Bi)</i>			
<i>FT (ref)</i>	0.154 (+2.27)	0.617 (+39.13)	0.403 (+10.73)
LoRA	-0.030 (-0.44)	0.406 (+25.72)	0.303 (+8.06)
LERP	0.239 (+3.52)	0.636 (+40.29)	0.240 (+6.38)
TIES	0.248 (+3.65)	0.701 (+44.43)	0.299 (+7.95)
<i>Whisper Large v3 (ED-MI)</i>			
<i>FT (ref)</i>	0.101 (+0.98)	0.709 (+28.84)	0.341 (+7.49)
LERP	0.323 (+3.13)	0.508 (+20.67)	0.190 (+4.18)
TIES	0.124 (+1.20)	0.721 (+29.31)	0.346 (+7.60)
<i>Qwen2.5-Omni-7B (ALLM-ASR-Bi)</i>			
<i>LoRA (ref)</i>	0.066 (+0.46)	0.227 (+7.86)	0.566 (+20.32)
LERP	0.070 (+0.49)	0.219 (+7.57)	0.643 (+23.08)
TIES	0.075 (+0.52)	0.215 (+7.43)	0.644 (+23.09)

Table 7: Child adaptation gain per benchmark. $\mathcal{AG}$ = Child Adaptation Gain on child sets, reported as relative WER reduction with Δ_{child} in WER points vs. θ_{base} . Higher $\mathcal{AG}$ and positive Δ are better. Within each architecture block, the best Δ per column is in blue.

For E-CTC, TIES is strongest, giving the highest $\mathcal{AG}$ on AraKidsL1 (0.248) and AraKidsL2 (0.701) and even exceeding full fine-tuning, while LoRA is the weakest and slightly degrades AraKidsL1 ($\mathcal{AG} = -0.030$).

For Whisper, LERP gives the best AraKidsL1 gain (0.323), whereas TIES is strongest on AraKidsL2 (0.721) and MyST (0.346).

For Qwen2.5-Omni-7B, child gains on the Arabic sets are small across methods, but both LERP and TIES clearly improve MyST over the LoRA reference ($\mathcal{AG}$ rising from 0.566 to ≈ 0.64). Overall, $\mathcal{AG}$ confirms that the strongest child-speech improvement comes from full fine-tuning for ED, but from TIES merging for E-CTC and Whisper.